

MATH 345 Skeleton Notes

Unit 2: Anatomy of a linear map

Section 2.1: Understanding a linear map: a preview

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In Vectors, matrices, and linear maps, a matrix defined a linear map

$$\mathbf{x} \mapsto A\mathbf{x}.$$

This unit answers the following questions about the behavior of this map. The second column previews the concepts introduced later in the unit for this purpose.

- **Question:** When do different inputs give the same output?
Concept/Method: $A(\mathbf{x} + \mathbf{z}) = A\mathbf{x}$
- **Question:** What input changes does A forget?
Concept/Method: $\text{null}(A)$
- **Question:** When can a desired output be hit?
Concept/Method: $A\mathbf{x} = \mathbf{b}$
- **Question:** What outputs can A produce?
Concept/Method: $\text{im}(A)$
- **Question:** Can the map be reversed?
Concept/Method: A^{-1} , when it exists

We give several preliminary examples.

Activity: A map that forgets height (U2-LO1, U2-LO3)

Let

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

First compute $P\mathbf{x}_1$, $P\mathbf{x}_2$, and $P\mathbf{x}_3$, where

$$\mathbf{x}_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 2 \\ 1 \\ 5 \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix}.$$

1. Which input coordinate changed?
2. Which part of the input did the output remember?
3. Which part of the input did the output forget?
4. Suppose the desired output is $(2, 1)$. What simple equations must the input coordinates satisfy?
5. Following up on the previous question, how many choices of the third coordinate work?

We compute

$$P \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1x + 0y + 0z \\ 0x + 1y + 0z \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}.$$

So the outputs are

$$P_{\mathbf{x}_1} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad P_{\mathbf{x}_2} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad P_{\mathbf{x}_3} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

The third coordinate changed, but the output did not. The map remembers the first two coordinates and forgets height.

To get the desired output $(2, 1)$, an input $\mathbf{x} = (x, y, z)$ must satisfy

$$x = 2, \quad y = 1.$$

There is no equation restricting z , so many (in fact, infinitely many!) inputs work:

$$\mathbf{x} = \begin{bmatrix} 2 \\ 1 \\ z \end{bmatrix}.$$

Activity: The same height loss in code (U2-LO3)

The following code repeats the same experiment.

```
import numpy as np

P = np.array([[1, 0, 0],
```

Activity: The same height loss in code (U2-LO3) (continued)

```
[0, 1, 0]])  
  
x1 = np.array([4, -2, 0])  
x2 = np.array([4, -2, 6])  
x3 = np.array([4, -2, -5])  
  
P @ x1, P @ x2, P @ x3
```

Predict the three code outputs.

The three outputs are $[4, -2]$, $[4, -2]$, and $[4, -2]$. The third coordinate changes across the inputs, but that change does not appear in the output.

Activity: Can this map produce the requested output? (U2-LO1, U2-LO3)

Let

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}.$$

First compute $A\mathbf{x}_1$, $A\mathbf{x}_2$, and $A\mathbf{x}_3$, where

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

1. Can this map produce the output $(1, 2, 3)$? If yes, give an input.

Activity: Can this map produce the requested output? (U2-LO1, U2-LO3)
(continued)

2. What rule do the outputs above all obey?
3. For the desired output (1, 2, 4), what equations would the unknown input $\mathbf{x} = (x, y)$ have to satisfy?
4. Following up on the previous question, can all of those equations be true at the same time?

We compute

$$A \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1x + 0y \\ 0x + 1y \\ 1x + 1y \end{bmatrix} = \begin{bmatrix} x \\ y \\ x + y \end{bmatrix}.$$

So the outputs are

$$A\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad A\mathbf{x}_2 = \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}, \quad A\mathbf{x}_3 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

The output (1, 2, 3) is produced by the input $\mathbf{x}_1$. The outputs above all obey the rule: third coordinate equals first coordinate plus second coordinate.

For the desired output (1, 2, 4), the unknown

input $\mathbf{x} = (x, y)$ would have to satisfy

$$x = 1, \quad y = 2, \quad x + y = 4.$$

The first two equations force $x + y = 3$, not 4. So these equations cannot all be true at the same time, and the output $(1, 2, 4)$ cannot be produced.

Activity: Two inputs, one output (U2-LO1, U2-LO6)

Let

$$F = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

First compute $F\mathbf{u}$ and $F\mathbf{v}$, where

$$\mathbf{u} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix}.$$

1. Are the inputs the same?
2. Are the outputs the same?
3. Specify $\mathbf{d}$ such that $\mathbf{v} = \mathbf{u} + \mathbf{d}$.
4. Check that $F\mathbf{d} = \mathbf{0}$.

The inputs are different:

$$\mathbf{u} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix}.$$

But the outputs are the same:

$$F\mathbf{u} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad F\mathbf{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

The change from $\mathbf{u}$ to $\mathbf{v}$ is

$$\mathbf{d} = \mathbf{v} - \mathbf{u} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}.$$

Writing $\mathbf{d} = (d_1, d_2, d_3)$, the condition $F\mathbf{d} = \mathbf{0}$ becomes

$$F\mathbf{d} = \begin{bmatrix} 1d_1 + 1d_2 + 0d_3 \\ 0d_1 + 1d_2 + 1d_3 \end{bmatrix} = \begin{bmatrix} d_1 + d_2 \\ d_2 + d_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

For the vector $\mathbf{d} = (1, -1, 1)$ found in item 3, we check both equations explicitly:

$$d_1 + d_2 = 1 + (-1) = 0,$$

$$d_2 + d_3 = (-1) + 1 = 0.$$

Therefore $Fd = 0$.

Activity: Same output in code (U2-LO6)

The following code compares two more inputs for the same map.

```
F = np.array([[1, 1, 0],
              [0, 1, 1]])

u = np.array([0, 1, 2])
v = np.array([1, 0, 3])
d = v - u

F @ u, F @ v, d, F @ d
```

Predict the four code outputs.

The four outputs are $[1, 3]$, $[1, 3]$, $[1, -1, 1]$, and $[0, 0]$. Thus $F @ u$ and $F @ v$ are the same, and $F @ d$ is $[0, 0]$.

The equation $F @ d = [0, 0]$ represents

$$d_1 + d_2 = 0, \quad d_2 + d_3 = 0.$$

The output does not always determine the input uniquely.

Activity: Output scores and a hidden direction (U2-LO6)

A final model layer can convert a hidden vector into output scores by

$$\ell = W\mathbf{h} + \mathbf{b}.$$

Let

$$W = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 0.5 \\ -0.5 \end{bmatrix}, \quad \mathbf{h} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{d} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}.$$

1. Compute $W\mathbf{h} + \mathbf{b}$.
2. Compute $W\mathbf{d}$.
3. Use part 2 to compute $W(\mathbf{h} + \mathbf{d}) + \mathbf{b}$.
4. Are $\mathbf{h}$ and $\mathbf{h} + \mathbf{d}$ the same hidden vector?
5. Are their output-score vectors the same?
6. Did the bias vector $\mathbf{b}$ affect which hidden-vector change was invisible?

First,

$$W\mathbf{h} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad W\mathbf{h} + \mathbf{b} = \begin{bmatrix} 2.5 \\ 0.5 \end{bmatrix}.$$

Next,

$$W\mathbf{d} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Therefore,

$$\begin{aligned} W(\mathbf{h} + \mathbf{d}) + \mathbf{b} &= W\mathbf{h} + W\mathbf{d} + \mathbf{b} \\ &= W\mathbf{h} + \mathbf{b} \\ &= \begin{bmatrix} 2.5 \\ 0.5 \end{bmatrix}. \end{aligned}$$

The hidden vectors $\mathbf{h}$ and $\mathbf{h} + \mathbf{d}$ are different, but they produce the same output scores. The change $\mathbf{d}$ is invisible to W because $W\mathbf{d} = \mathbf{0}$.

The bias vector shifts the output scores, but it does not affect which hidden-vector changes are invisible.

Here is the same issue in a data-matrix setting.

Activity: Two coefficient choices, same predictions (U2-LO1, U2-LO6)

A data matrix stores three measured features for four observations:

$$X = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 1 \\ 1 & -1 & 0 \end{bmatrix}.$$

Observe that the third feature is the sum of the first two features in every row.

Consider the two coefficient vectors

$$\mathbf{c} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{c}' = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

In this example, the linear map takes a coefficient vector $\mathbf{c}$ as input and returns the prediction vector $X\mathbf{c}$.

1. Compute $X\mathbf{c}$.
2. Compute $X\mathbf{c}'$.
3. Are the coefficient vectors the same?
4. Are the predictions the same?
5. Let $\mathbf{d} = \mathbf{c} - \mathbf{c}'$. What equations does $X\mathbf{d} = \mathbf{0}$ represent?

We have

$$X\mathbf{c} = \begin{bmatrix} 4 \\ 8 \\ 1 \\ 1 \end{bmatrix}, \quad X\mathbf{c}' = \begin{bmatrix} 4 \\ 8 \\ 1 \\ 1 \end{bmatrix}.$$

The coefficient vectors are different, but the predictions are the same.

The difference is

$$\mathbf{d} = \mathbf{c} - \mathbf{c}' = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}.$$

The equation $X\mathbf{d} = \mathbf{0}$, with $\mathbf{d} = (d_1, d_2, d_3)$, represents

$$\begin{aligned} d_1 + 2d_2 + 3d_3 &= 0, \\ 2d_1 + 4d_2 + 6d_3 &= 0, \\ d_2 + d_3 &= 0, \\ d_1 - d_2 &= 0. \end{aligned}$$

The vector $(1, 1, -1)$ satisfies these equations.

This happens because the third feature repeats information already contained in the first two features: feature 3 equals feature 1 plus feature 2 in every row.

Each preview led to the same kind of question: can certain linear equations in the unknown input coordinates be solved? The next section gives this question a formal name and a systematic method.